

Video And Image Manipulation Detection

A Project Report

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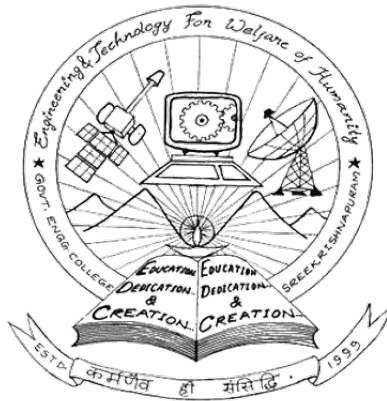
in partial fulfilment of requirements for the award of degree

of

Bachelor of Technology

in

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CERTIFICATE

This is to certify that the report entitled **Video And Image Manipulation Detection** submitted by **ALWIN JOSEPH** (PKD19CS008), **MUHAMMED RIZAL K E** (PKD19CS036), **SANKAR VINAYAK E P** (PKD19CS046), **SHIBIL AHAMMED M** (PKD19CS050), to the APJ Abdul Kalam Technological University in partial fulfillment of the B.Tech. degree in Computer Science and Engineering is a bonafide record of the project work carried out by him under my guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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We ALWIN JOSEPH, MUHAMMED RIZAL K E, SANKAR VINAYAK E P, SHIBIL AHAMMED M hereby declare that the project report **Video And Image Manipulation Detection** , submitted for partial fulfilment of the requirements for the award of the degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under supervision of Liji L Dominic

This submission represents my ideas in my own words and where ideas or words of others have been included, We have adequately and accurately cited and referenced the sources.

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Abstract

Video and image manipulation, such as deepfakes, and AI-generated and computer-generated media, has become a major concern in recent years due to the advancements in technologies. With the help of these, it is possible for misusing visual media for malicious purposes. Detecting manipulation in videos and images is crucial for various applications, including forensics, journalism, and social media. In this project, we address the problem of detecting manipulated and generated videos and images using deep learning techniques, with a focus on methods relevant to forensic analysis. We explore and analyze existing methods and technologies for detecting deepfakes and CGI, manipulations, and develop and test novel approaches to arrive at a robust and effective solution. Our evaluation of the performance of different algorithms demonstrates the feasibility and effectiveness of our proposed methods in detecting various types of manipulations. The results of this project have important implications for forensic analysis and other applications where the authenticity of visual media are crucial. The development of reliable methods for detecting manipulations using deep learning techniques can help ensure the credibility and trustworthiness of visual media in various contexts.

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Abbreviation

Abbreviation	Description
CFFN	Common Fake Feature Network
ANN	Artificial Neural Network
NN	Neural Network
SVM	Support Vector Machine
DFT	Discrete Fourier transforms
CGI	Computer generated image
ViT	Vision transformer
GTM	Ground Truth Mask
MFR	Median Filter Residual
RNN	Recurrent Neural Network
CNN	Convolutional neural networks
LSTM	Long Short Term Memory
DFDC	Deepfake detection challenge
MTCNN	Multi-Task Cascaded Convolutional Neural Network
LSTM	Long Short-Term Memory

Chapter 1

Introduction

1.1 General background

Video and image manipulation has become a major concern in recent years due to the widespread use of digital media and advances in editing software. The ability to manipulate and generate visual media can have significant impacts on a variety of fields, including forensics, journalism, and social media. As such, detecting manipulation in videos and images has become an important research topic.

Existing methods for detecting manipulation in videos and images include traditional forensic analysis techniques, such as error-level analysis and image quality assessment, as well as newer approaches using machine learning and computer vision. Deep understanding, in particular, is effective for detecting manipulation in images and videos due to its ability to learn features and patterns from data.

Chapter 2

Literature Review

2.1 Exposing Manipulated Photos and Videos in Digital Forensics Analysis

Introduction

The rise of tampered multimedia content has created a new challenge for a criminal investigation in the age of digital crime. With the increasing use of manipulated digital photos and videos, cybercriminals are spreading fake news, misinformation, digital kidnapping, and ransomware-related crimes. It has become crucial to apply machine learning techniques to automatically distinguish between fake and genuine seized multimedia content. However, manual validation is still necessary in many cases. Therefore, easy-to-use platforms for digital forensics are essential to automate and facilitate the detection of tampered content, and to aid criminal investigators in their work. In this context, this article will discuss the challenges of detecting tampered multimedia content and the potential solutions to automate the detection process using machine learning techniques.

Methodology

This paper [2] presents a novel machine learning-based approach for distinguishing between genuine and fake multimedia files, including digital photos and videos. The method uses Support Vector Machines (SVM) and applies a Discrete Fourier Transform (DFT) to extract a set of simple features from the multimedia content.

The authors implemented the approach in Python and integrated it into the digital forensics application Autopsy. They evaluated the model with a large dataset of multimedia files containing both genuine and fake content, including deepfake videos from the Celeb-DFv1 dataset. The results of the 5-fold cross-validation showed that the proposed method outperformed existing SVM-based methods documented in the literature, achieving an average F1-score of 99.53% for photos, 79.55% for videos, and 89.10% for a mixture of both types of content. Although the proposed DFT-SVM approach was not as competitive as deep learning approaches like Convolutional Neural Networks (CNN), it had a substantially reduced processing time and was appropriate for implementation and embedding into Autopsy modules for predicting the level of the fakeness of each analyzed multimedia file. Overall, this study presents a promising approach for detecting deepfake content and can provide valuable support for digital forensics investigations.

Advantages

The advantages of this approach include higher accuracy since it is performing well on different sets of datasets. The method uses simple features extracted using DFT, which allows for a reduced computational cost compared to more complex deep learning methods. The method was integrated into the Autopsy digital forensics application, making it easy to use for criminal investigators.

Disadvantages

There are some issues with this model including the method uses SVM as the machine learning algorithm, which may not be as powerful as other methods such as deep learning approaches like CNNs, this method relies on simple features extracted using DFT, which may not capture all the nuances of fake multimedia content, particularly more sophisticated deepfakes. Also, it can't handle other types of fake content which are highly realistic and completely generated like AI-generated and computer graphics.

2.2 Detecting Computer Generated Images with Deep Convolutional Neural Networks

Introduction

Computer-generated images are becoming increasingly photo-realistic due to the increased computational power. The help of advanced image generation and simulation techniques makes it difficult for humans to differentiate between computer-generated and photographic images. But the machine learning techniques can detect small anomalies based on pixel values which can be used to differentiate them.

Methodology

for the detection of computer-generated images with deep convolutional neural network [3] the researchers used ResNet50 a popular image classification model for the purpose of feature extraction. They replaced the classification layer of ResNet50 with a support vector machine (SVM) binary classification layer. After passing the image through the feature extraction layer it is passed through the layer which extracts the bottleneck features which have much lower dimensions than the input image. Based on the extracted bottleneck features the classifier classifies the images into computer-generated and photographed. There are no changes except for the transforms for ResNet are used as preprocessing phase, figure 2.1 shows the architecture of the model.

Advantages

By using deep convolutional neural networks (CNNs) and support vector machines (SVMs) for image classification, the researchers may have been able to achieve high levels of accuracy in detecting computer-generated images. CNN and SVM are both relatively robust to variations in the data and are able to handle complex patterns and features, which may make the approach effective at detecting computer-generated images that show subtle differences from real photographs. By using transfer learning to fine-tune a pre-trained CNN model, the researchers may have been able to significantly reduce the amount of data and computation required for training the

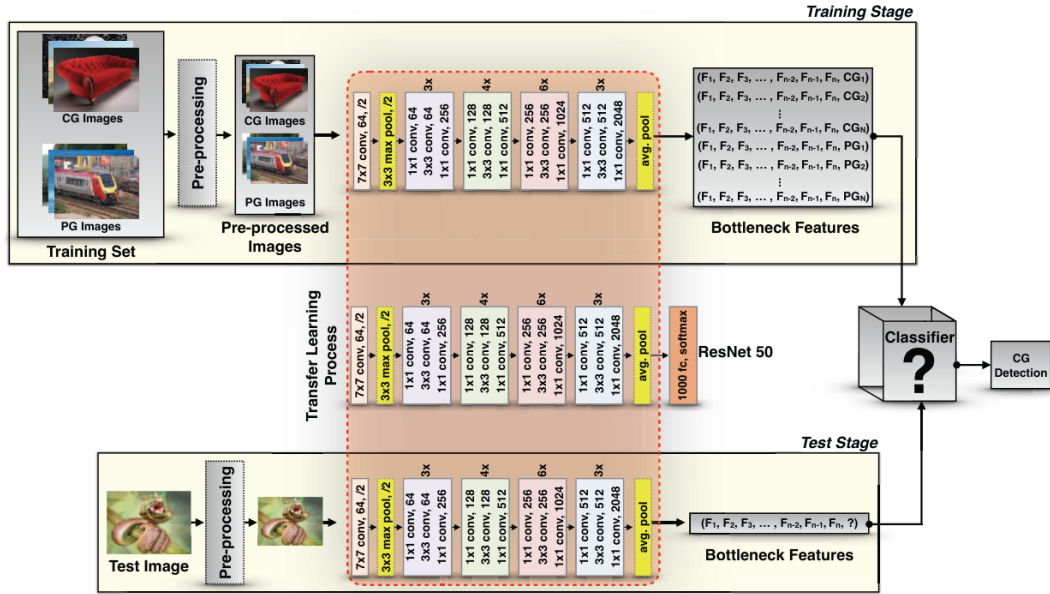


Figure 2.1: Architecture of CGI image detection

model, making the approach more efficient and practical to implement. The approach may be able to classify the images without any preprocessing, which could make it easier to implement and more flexible in terms of the types of images it can handle.

Disadvantages

The effectiveness of the approach may depend on the specific techniques used to create the computer-generated images. If the images are created using techniques that are not represented in the training data, the model may not be able to detect them accurately. The quality and diversity of the training data can significantly impact the performance of the model. If the training data is not representative of the types of computer-generated images that the model will encounter in real-world scenarios, the model may not perform well. The model may only be effective for detecting computer-generated images that are similar to those in the training data. It may not be able to detect images that have been created using entirely new techniques or that show significant differences from the training data. The model may be complex and require a significant amount of computation and data to be trained and tested, which could make it difficult to implement and maintain. The model may rely on machine learning techniques, which can be sensitive to variations in the data and may not always be reliable. There is also a risk that the model could be fooled by computer-generated images that are

particularly well-made or by images that have been created using new techniques that were not seen in the training dataset.

2.3 An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale

Introduction

This paper [1] proposes a novel approach to image classification using a transformer-based architecture called Vision Transformer (ViT). The paper presents an evaluation of the ViT model on various image classification benchmarks, demonstrating its superior performance compared to existing state-of-the-art CNN-based models. The ViT model has since garnered significant interest in the deep learning community and has been the subject of further research exploring its potential applications and improvements. Therefore, this paper serves as a crucial reference in any report discussing ViT and its applications in image recognition

Methodology

The ViT model divides the input image into a grid of non-overlapping patches, which are then flattened and linearly projected to obtain a sequence of patch embeddings. The patch embeddings are then fed into a series of transformer encoder layers, where self-attention mechanisms are applied to capture the dependencies between different patches. The output of the transformer encoder layers is then fed into a linear classification head, which predicts the class label of the input image.

Advantages

The ViT model offers several advantages over traditional convolutional neural network (CNN) models. One key advantage is its better performance on certain image classification benchmarks, such as ImageNet. Additionally, the ViT model has better-scaling properties than traditional CNN models, making it well-suited for large-scale image classification tasks. The self-attention mechanism used in the ViT model also

makes it easier to interpret and visualize the features that the model has learned, improving our understanding of how the model makes its predictions. Finally, the ViT model's ability to be pre-trained on large datasets using self-supervised learning techniques can improve its generalizability to new datasets and tasks.

Disadvantages

While the ViT model offers several advantages, it also has some disadvantages. One major disadvantage is its high computational requirements, requiring large amounts of memory and processing power, which can make it difficult to train or deploy on resource-constrained devices or systems. Additionally, the ViT model is primarily designed for image classification tasks and may not be as effective for other computer vision tasks, such as object detection or segmentation. Finally, while the self-attention mechanism used in the lower layers of the ViT model makes it easier to interpret the learned features, the higher layers of the model may be more difficult to interpret, making it challenging to understand how the model makes its predictions at a higher level of abstraction.

2.4 Deepfake Video Detection Using Recurrent Neural Networks

Introduction

The use of convolutional autoencoders and generative adversarial networks (GANs) has made it easier for individuals to tamper with images and videos. Tools such as FaceApp and FakeApp use this technology to allow users to alter facial attributes or create manipulated videos called "deepfakes". To detect deepfake videos, a two-stage analysis method was presented which combines a CNN for feature extraction at the frame level with a temporally-aware RNN network to identify temporal inconsistencies introduced by the face-swapping process. This method was tested on a collection of 600 videos, half of which were deepfakes.

Methodology

The presented system for detecting deepfake videos uses a convolutional LSTM to analyze frame sequences. The convolutional LSTM has two main components: a CNN that extracts features from individual frames and an LSTM that analyzes the temporal sequence of frames. When presented with an unseen test sequence, the system first extracts features from each frame using the CNN and concatenates the features of multiple consecutive frames. The concatenated features are then passed to the LSTM for analysis, which produces an estimate of the likelihood that the sequence is a deepfake or a non-manipulated video. This process allows the system to effectively detect deepfake videos.

The system uses a pre-trained InceptionV3 network to extract features from the input frames, which are input to the LSTM for temporal sequence analysis. The InceptionV3 network is not fine-tuned for this task and is used with its pre-trained weights. The LSTM unit has a 2048-dimensional width and a 0.5 chance of dropout, and is followed by a fully-connected layer with a 512-dimensional width and a 0.5 chance of dropout. The LSTM module is trained end-to-end without the need for auxiliary loss functions. A shallow network consisting of two fully-connected layers and one dropout layer is also used to map the high-dimensional LSTM descriptor to a final detection probability and minimize training over-fitting.

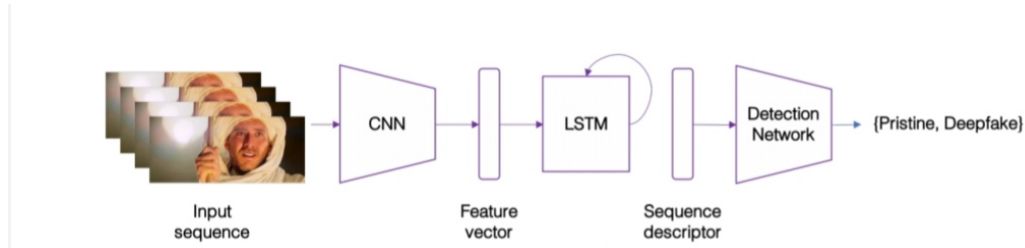


Figure 2.2: Overview of detection system

Advantages

The authors of the paper [7] presented a system for detecting deepfake videos using a convolutional LSTM structure. The system analyzes the temporal sequence of frames to identify inconsistencies introduced by the deepfake process. The authors conducted

experiments on a large collection of manipulated videos . One of the strengths of the system is its simplicity, as it has a straightforward pipeline architecture and is easy to implement and maintain. In the future, the authors plan to improve the system's robustness against manipulated videos using unseen techniques during training. The system is also able to detect deepfakes in short video clips, as it can accurately predict whether a video has been subject to manipulation using as little as 2 seconds of video data.

Disadvantages

The authors of the paper presented a temporal-aware system for detecting deepfake videos using a simple convolutional LSTM structure. The system was tested using an extensive collection of manipulated videos and was able to accurately predict whether a video had been subject to manipulation with as little as 2 seconds of video data. The authors claim that their system provides a strong first line of defence against fake media created using AI techniques and demonstrated competitive results in this task. However, the system has some limitations, including its inability to detect certain types of deepfakes and its sensitivity to low-quality or heavily distorted input. Additionally, it is specifically designed to detect deepfake videos and may not be effective at detecting other types of manipulated media. The training process also requires an extensive collection of manipulated videos and a strong computational setup. In future work, the authors plan to investigate ways to increase the system's robustness against manipulated videos using unseen techniques during training.

2.5 Detection of Spliced Image Forensics Using Texture

Analysis of Median Filter Residual

Introduction

It is important to be able to determine whether an image has been manipulated, as manipulated images can be used for deceptive or fraudulent purposes. For example, manipulated images can be used to spread misinformation or propaganda, to deceive

people into believing something that is not true. In this way, manipulated images can have serious consequences, as they can be used to influence public opinion or to shape people's perceptions of events.

Image manipulation detection is important in the field of image forensics, where it is used to identify and analyze images that have been manipulated or altered in some way. This can be useful for a variety of purposes, including identifying forged documents or detecting tampering with evidence.

In this paper, for the detection of spliced images, a new Image Forensics Detection (IFD) scheme is presented. In this scheme the feature vector is extracted from the texture analysis of MFR. Thus, the suggested algorithm has the characteristics of image filtering process and texture analysis.

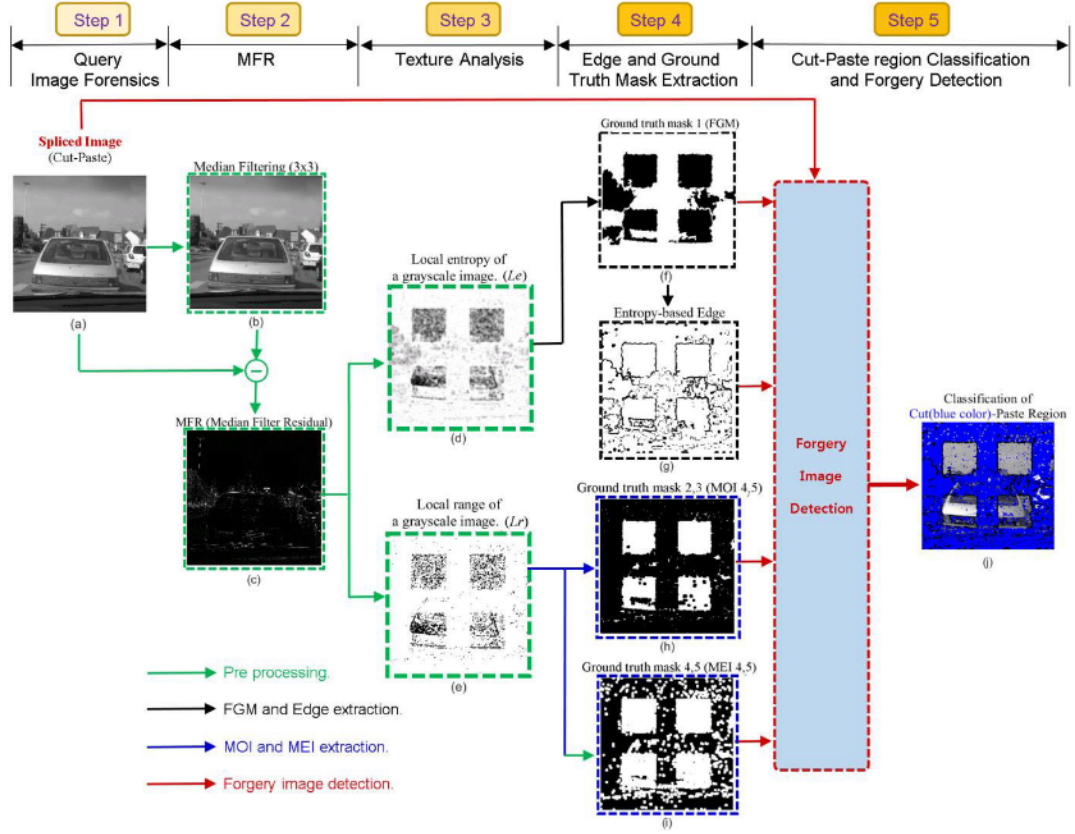


Figure 2.3: Architecture of Spliced Image Detection

Methodology

The suggested scheme contains five steps. Begin by providing an input which is a forged/spliced image which has Cut and Paste region. Then Median filter is applied to the input image. The median filter residual is then found by using the difference between the input image and the median filtered image. Texture analysis of MFR is done to get two images i.e. local entropy and local range. The local entropy(Le) and Local range(Lr) is formulated as given below:

$$Le = Ef(MFR)$$

$$Lr = Rf(MFR)$$

where Ef is the entropy filter and Rf is the range filter. Now, the generation of five ground truth masks (GTM) to classify the Cut-Paste region in the spliced images Le (d) and Lr (e). The GTM feature set is extracted using morphological open (mo) and erosion (me) operations.

The FGM (Find Gray level regional Maxima of closing by image reconstruction) process is represented by the black dashed line (f) and consists of the following steps:

$$Ie = me(Lr, r), \text{ where } Ie \text{ is the morphological erosion image.}$$

$$Ir = rec(Ie, Lr), \text{ where } Ir \text{ is the reconstruction image.}$$

$$Id = rec(Ir, r), \text{ where } Id \text{ is the reconstruction image.}$$

Here, r is the radius in pixels for the morphological operation, and rec(·) represents the reconstruction image. In addition to the morphological open (MO) and erosion (ME) operations, two morphological-open images (MOI) and two morphological-erosion images (MEI) are also extracted using mo and mi. The blue represents these four morphological-based GTMs (h, i) dashed lines. The MOI and MEI are formulated as follows:

$$MOI(r) = mo(Le, r), \text{ where } r = 4 \text{ or } 5.$$

$$MEI(r) = me(Lr, r), \text{ where } r = 4 \text{ or } 5.$$

Finally, the forgery image detection block (represented by the red dashed line) uses five input elements (a, f, h, and i) to classify the Cut-Paste region (j) in the spliced

image (a). The resulting detection image (j) includes a Cut region (shown in blue), a Paste region (represented by the grayscale image itself), and a Cut-Paste border edge (denoted by the thick black line).

Advantages

The paper presents a method for detecting spliced images, which are images that have been manipulated by combining two or more images together. The method uses texture analysis of the median filter residual, which is the difference between the original image and a median-filtered version of the image.

There are several advantages to this approach:

1. It is effective at detecting spliced images: The method has been shown to have high accuracy in detecting spliced images in experiments.
2. It is robust to image processing: The median filter residual is relatively robust to image processing, such as resizing or compressing the image, so the method can still detect spliced images even if they have been processed.
3. It is efficient: The method is computationally efficient, making it practical to use on large datasets or in real-time applications.
4. It is easy to implement: The method is based on simple image processing techniques, making it easy to implement and replicate.

Overall, the method presented in the paper "Detection of Spliced Image Forensics Using Texture Analysis of Median Filter Residual" is a useful tool for detecting spliced images and has a number of practical advantages.

Disadvantages

One potential disadvantage of the method is that it may not be effective at detecting all types of spliced images. The method is based on analyzing the texture of the median filter residual, which may not be sensitive to all types of image splicing. For example, if the spliced region of the image has similar texture to the rest of the image, the method may not be able to detect the splice.

Another potential disadvantage is that the method may produce false positives, where it incorrectly identifies a non-spliced image as a spliced image. This could be due to the presence of texture patterns in the image that are similar to those produced by splicing.

2.6 An approach based on object detection for image forensics

Introduction

Digital image manipulation and counterfeiting have become widespread issues with the proliferation of the internet and advanced image editing software. As a result, research in the field of image forensics has become increasingly important. In this paper, the authors propose an approach based on the Faster R-CNN network, a powerful object detection tool, for detecting digital image forgery. The approach uses two channels for extracting features from the image. One channel focuses on detecting noise inconsistencies between real and tampered areas, while the other channel, based on the original Faster R-CNN network, looks for evidence of tampering such as strong contrast differences and unnatural tampering boundaries. The features from these two channels are then fused through a bilinear collection layer to further identify tampering traces. Experiments on four standard image tamper datasets show that this two-channel approach outperforms single detection networks and has better performance than previous methods that were able to handle resizing and compression.

Methodology

The paper presents a method for detecting digital image forgery using the Faster R-CNN network, a powerful object detection tool.

The method uses two channels for extracting features from the image. The first channel is designed to detect noise inconsistencies between real and tampered areas, while the second channel, based on the original Faster R-CNN network, looks for evidence of tampering such as strong contrast differences and unnatural tampering

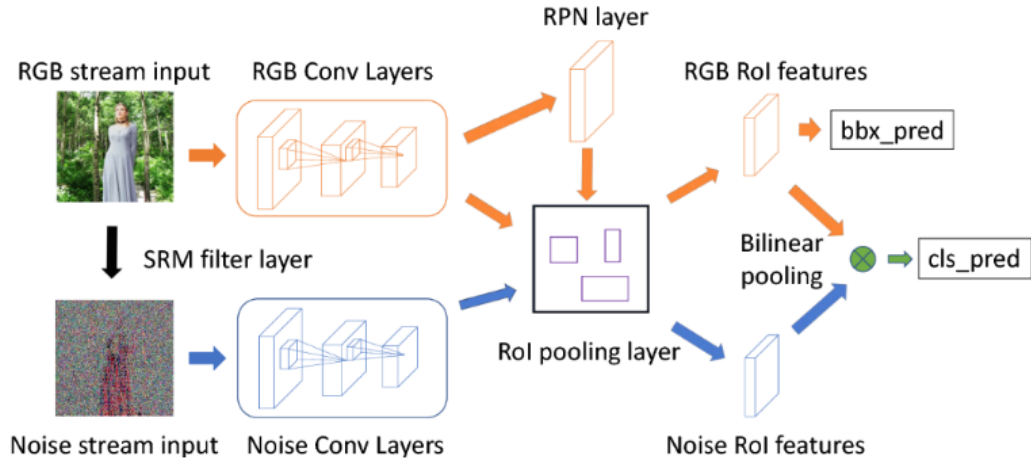


Figure 2.4: Architecture of Object Detection

boundaries.

To detect noise inconsistencies, the first channel adds an additional layer to the Faster R-CNN network specifically designed to extract noise features. The second channel uses the original Faster R-CNN network to extract features from the original image.

The features from these two channels are then fused through a bilinear collection layer to further identify tampering traces. The fused features are then passed through a fully connected layer and a softmax layer to classify the image as either forged or authentic.

Finally, the method combines the use of the Faster R-CNN network with additional layers designed to detect noise inconsistencies and tampering evidence to effectively identify digital image forgery.

Advantages

The advantages of this method are:

1. It is effective at detecting forgery: The method has been shown to have high accuracy in detecting forged images in experiments.
2. It is efficient: The Faster R-CNN network is known for its computational efficiency, making the method practical to use on large datasets or in real-time applications.

3. It is robust to image processing: The method is relatively robust to image processing, such as resizing or compressing the image, so it can still detect forgeries even if the image has been processed.
4. It is flexible: The method can handle a wide range of forgery types, including tampering with the content of the image, such as adding or removing objects, as well as tampering with the appearance of the image, such as adjusting the contrast or colour.

Disadvantages

The method is based on the Faster R-CNN network, which is a powerful object detection tool, but may not be sensitive to all types of image manipulation.

Another potential disadvantage is that the method may produce false positives, where it incorrectly identifies a non-forged image as a forged image. This could be due to the presence of features in the image that are similar to those produced by forgery.

Chapter 3

Proposed system

Our proposed system will help in handling these different types of manipulations with the help of a deep-learning base model. The overall working of the model and its training process is described in figure 3.1

Different techniques have to be used for the detection of Deepfake videos, manipulated and generated image detectors.

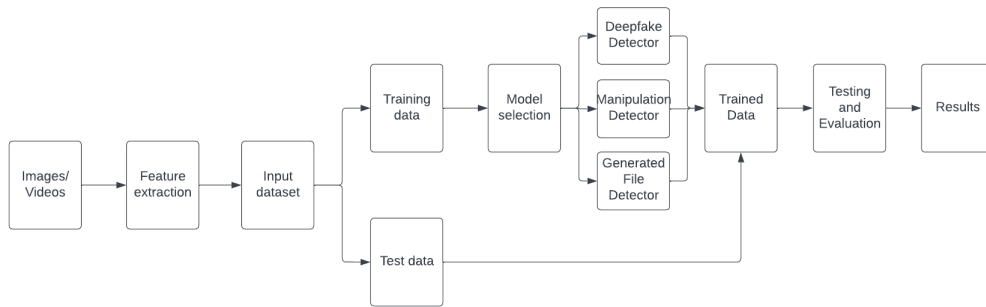


Figure 3.1: Block Diagram of the methodology

The data will be collected from a combination of standard datasets and files available freely on the internet. A simple user interface will be provided for easy usage of the different models for the end user.

Preprocessing files before giving them to the respective models will be handled by the UI.

3.1 Feasibility study

Technical feasibility

For training, there are a variety of datasets available which have free or permissioned access. Also by making use of the freely available data over the internet help in creating the dataset.

There are a variety of libraries which can assist in the development of this project like PyTorch which is based on the torch framework, and Tensorflow are available. They help in reducing the complexity associated with the creation of deep neural networks.

For the user interface, we can use any framework like Django, Flask, etc or there are libraries like Gradio and Streamlit which make creating UI for ML applications much easier.

Financial feasibility

For the purpose of training and deployment, specific hardware devices such as cuda enabled GPU are required. There are services such as Google Collab, AWS sagemaker studio lab, Paperspace gradient are available at free or low cost. For deployment services like Hugging face spaces provide an environment which satisfies the requirements. By using these services projects can be done at low cost.

Chapter 4

Preliminary analysis

4.1 Software requirements

- **Python:** Python is a high-level programming language known for its simplicity and readability. It provides a wide range of libraries and frameworks for various purposes, including scientific computing, web development, and machine learning.
- **face-recognition:** The face-recognition library is a powerful tool for face detection and recognition tasks. It provides a simple interface to locate and manipulate faces within images or video streams. The library utilizes deep learning models to identify and classify faces based on their unique features.
- **PyTorch:** PyTorch is a popular open-source machine learning framework that facilitates the development and training of neural networks. It offers dynamic computation graphs and a wide range of built-in functions, making it flexible and efficient for building various deep-learning models.
- **torchvision:** The torchvision library is a part of PyTorch and provides a collection of image and video datasets, pre-trained models, and various image transformation utilities. It simplifies the process of working with image data in PyTorch and offers convenient tools for data augmentation, model evaluation, and more.
- **streamlit:** Streamlit is a Python library specifically designed for creating interactive and customizable web applications for machine learning and data

science. It allows developers to build intuitive user interfaces with minimal code, making it easier to showcase and deploy machine learning models and visualizations.

- **sklearn:** Scikit-learn, often referred to as sklearn, is a widely used machine learning library in Python. It offers a comprehensive collection of algorithms and tools for data preprocessing, feature selection, model training, and evaluation. Scikit-learn is known for its user-friendly interface and extensive documentation, making it a popular choice for both beginners and experienced machine learning practitioners.

4.2 Hardware requirements

- CPU: i5-5200U/Ryzen 5 2500U or newer
- NVIDIA GPU with CUDA: GTX 1060 2GB or newer
- 8GB ram
- 5GB storage
- Latest version of a modern browser:firefox,chromium

Chapter 5

Methodology

There are various methods that can be used to manipulate or create media files. Based on the techniques used in creating or manipulating a given multimedia file different methods may need for the proper identification of these.

5.1 Deepfake detector

Deepfakes are a type of synthetic media that involve the replacement or manipulation of a person's likeness in an existing image or video. This is accomplished through the use of deep learning algorithms that analyze and replicate patterns found in the original data, allowing for the creation of realistic-looking videos or images that appear to feature someone else's likeness. Deepfakes are a growing concern in the digital world because they have the potential to be used to spread false information, manipulate public opinion, or even perpetrate fraud and other types of crimes, so deepfake detection is important.

For the model training, we used Celeb-DF [13] dataset which contains both real and fake videos of celebrities. The dataset is designed for research in deepfake detection and the videos for the dataset were collected from YouTube and cover a wide range of topics, including interviews, speeches, and public appearances.

For preprocessing the videos were split into frames and face detection was performed on each frame. The frames were then cropped to focus solely on the detected facial region. This approach was taken because facial features are where irregularities are most likely to occur in deepfake videos. since the original videos have a high number of frames and take lots of computation power and time, The number of

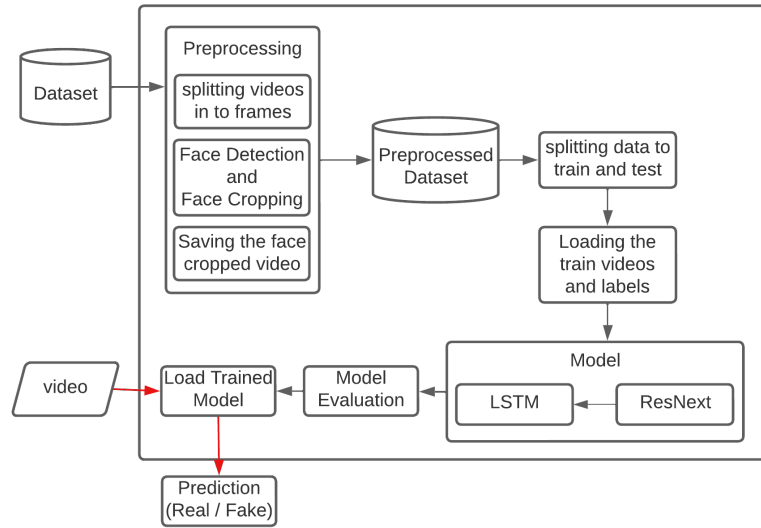


Figure 5.1: Model architecture

frames was reduced to 150 frames and the frames that don't have faces were ignored. Also the resolution was reduced to 122 x 122.

The model network consists of three main components, a convolutional neural network (CNN) for feature extraction, a long short-term memory (LSTM) network for temporal modeling and a fully connected layer for classification. The CNN used in this architecture is the ResNext50_32x4d model, which is pre-trained on the ImageNet dataset and has been shown to perform well on various computer vision tasks. The pre-trained weights are used for the CNN layers.

The output of the CNN is passed through an adaptive average pooling layer that reduces the spatial dimensions to 1x1 and averages the feature maps across all spatial locations. This results in a fixed-length feature vector that captures the salient features of the input frames. The feature vector is then fed into an LSTM network which is responsible for modeling the temporal dynamics of the video.

The output of the LSTM is passed through a dropout layer to prevent overfitting and then fed into a fully connected layer. The output of the fully connected layer is compared to the ground-truth labels to compute the loss.

5.2 Synthetic media detector

Computer-generated images these days became increasingly photo realistic due to the advancements in rendering technologies like Unreal Engine 5.2, octane, Blender, real-time ray tracing, etc. Using these and good 3-D modelling will result in highly realistic images. By the use of generative adversarial networks and transformers like Midjourney, Dalle-2, Stable diffusion and many more models these days we can easily convert our imagination into images without much skill. As their popularity increase it became increasingly complex to differentiate one another.

The CGI dataset is a combination of Columбина state university provided and the newer images available on the internet. For the AI images since there are no standard datasets available yet all images are sourced from the internet. For photographs, a combination of images from various sources is used.

The method we used here is based on the Vision Transformer [1]. It is a transformer-based architecture used in image classification and speech recognition applications. The weights of ViT are based on the values available after training it on the ImageNet dataset. In order to reduce the computational requirements the image gets resized into 224×224 before passing into the network. Also, the pixel value's mean and standard deviation are shifted which helps in better generalisation and reducing computation. The architecture of ViT contains many stages

- Patch + Position Embedding (inputs) - Turns the input image into a sequence of image patches and adds a position number in what order the patch comes in.
- Linear projection of flattened patches (Embedded Patches) - The image patches get turned into an embedding, the benefit of using an embedding rather than just the image values is that an embedding is a learnable representation (typically in the form of a vector) of the image that can improve with training.
- Layer Normalization - a technique for regularizing (reducing overfitting) a neural network
- Multi-Head Attention - Multihead self-attention layer

- Multi-layer perceptron - a collection of feedforward layers also referred to as MLP block
- Transformer encoder - A collection of the layers mentioned above

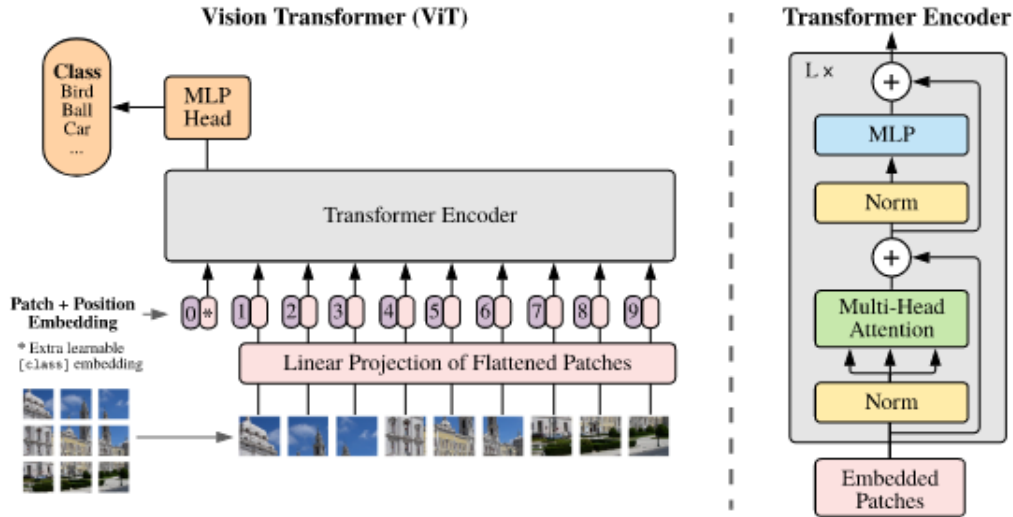


Figure 5.2: ViT architecture

The MLP Head consists of layers which take the input from the ViT output and converts it to lower dimensional output before passing through the non-linear activation function. Here the output from ViT which is of shape 768 is reduced to 256. It is then converted to the required output dimension of three, all by using a fully connected feed-forward neural network. The non-linear activation before the output layer in our case rectified linear unit is used for improving the prediction confidence and eliminating the connections which do not affect the final results. All the weights values except for the MLP head are frozen during the training process and the MLP head is trained to fit the requirement. For improving the performance of the model an optimal base learning rate is set as 0.001 and the batch size for training the model is set to 64. The learning rate was set to reduce by 0.5 if the loss does not improve over three epochs.

5.3 Manipulation detector

Image alteration is now simpler than ever because of the recent growth of digital photos and the widespread use of gadgets like smartphones and tablets. Furthermore, the

general public now has even greater access to this activity thanks to the low cost of picture modification software. As a result, some photographs are altered so skillfully that even humans are unable to tell the difference.

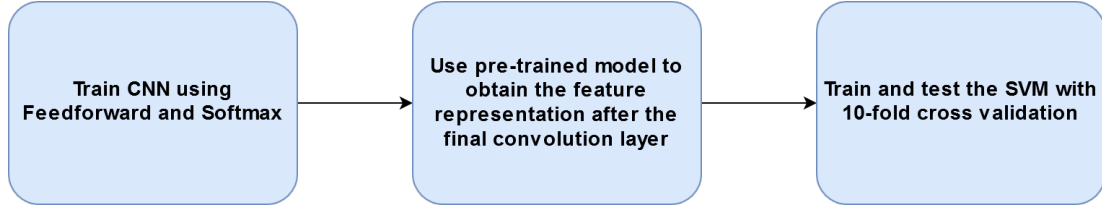


Figure 5.3: Pipeline

The main objective is to develop a pipeline that can differentiate between manipulated and genuine images. To achieve this goal, inspiration was drawn from Y. Rao et al's [10] proposed architecture, which employs a CNN as a feature extractor that takes an image patch $\mathbf{X} \in \mathbf{R}^{p \times p}$ as input and generates a feature representation $\mathbf{Y} = f(\mathbf{X}) \in \mathbf{R}^K$, where K represents the number of dimensions.

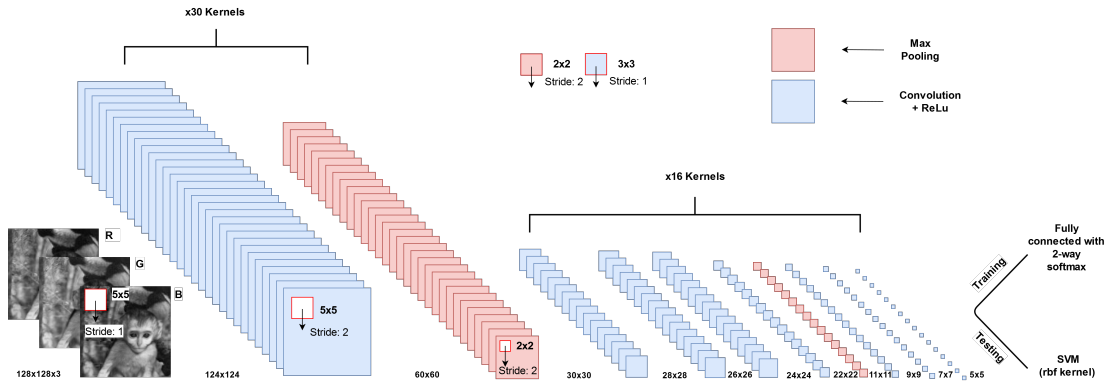


Figure 5.4: CNN model

The resulting feature representation is then processed by an SVM classifier to determine whether the input image is authentic or manipulated.

CNNs, which are members of the deep neural network family, are mostly used for image processing. Multiple convolutional layers, fully connected layers, and a softmax classifier make up the basic structure of a CNN. The input and output of each convolutional layer are referred to as feature maps. A convolutional layer is made up of a convolution, a non-linear activation function, and a pooling operation.

The CNN utilized (Fig.5.4) comprises nine convolutional layers and two max-pooling layers, with the input size of the network being a 128x128x3 patch, where

3 indicates the RGB colour channels. The first two convolutions have a kernel size of 5x5 and output 3 and 30 kernels, respectively, followed by a pooling operation using a 2×2 filter. The subsequent eight layers have 16 kernels, with convolutions and max pooling using 3x3 kernels and 2x2 filters, respectively. All convolutional layers use the ReLU activation function, and local response normalization is applied to every feature map before the pooling operation to enhance generalization [10].

To train the CNN, image patches with a size of 128x128x3 are extracted from the dataset with the help of a mask to give the CNN architecture the capacity to focus on the local areas of the artefacts and learn to detect them. The patches are extracted using the aforementioned method and then entered into the CNN. A 400-dimensional (5x5x16) feature representation of the patches is created by the CNN and fed into a fully-connected layer afterwards. This layer employs a dropout-based 2-way softmax classifier [11].

The SVM classifier uses this feature vector to differentiate between genuine and altered images after the CNN extracts the 400-dimensional feature representation of the image patches.

The pipeline (Fig.5.3) is implemented in Python 3.10 using PyTorch 2.0 to implement the CNN and the SVM3 [12] implementation provided by sci-kit learn.

5.4 User interface

The UI provided is a simple tool that can be used for image and video manipulation detection. The tool allows the user to upload an image or a video and select from Deepfake, CGI, and Splicing or Copy move. Based on the user's selection, the tool will display the output indicating whether the uploaded file is manipulated or not.

To use the tool, the user needs to upload an image or a video file in one of the accepted formats (jpg, jpeg, png, mp4, avi). The user can then select from the three options using the radio buttons. Once the user clicks the submit button, the tool checks the file type and displays the uploaded image or video. Then, depending on the selected option, the tool displays either "Deepfake," "CGI-AI" or "Copy move/Splicing" as the result.

The provided UI implements a simple user interface using Streamlit, a Python

framework for building web applications. Streamlit is a powerful Python library used to create interactive web applications. It simplifies the process of building and deploying data-driven applications by providing an intuitive and easy-to-use API.

The UI allows users to upload an image or video file and select from different options to detect image and video manipulation. The UI works as follows:

- **File Upload:** Users can choose an image or video file from their local system using the file uploader.
- **Option Selection:** Based on the file type, users can select the appropriate option using the radio buttons. This selection determines the type of manipulation detection to be performed.
- **Processing:** The uploaded file is processed based on the selected option. For images, any image processing or manipulation detection algorithms can be implemented using the provided image data. For videos, similar techniques can be applied frame by frame.
- **Result Display:** After the processing is complete, the result is displayed with a larger font size and the heading 'RESULT'. The result corresponds to the selected option and indicates whether the uploaded file exhibits the selected type of manipulation or not.

The UI components are:

- **File Uploader:** The UI includes a file uploader widget that allows users to upload image or video files. The accepted file formats are jpg, jpeg, png, mp4, and avi.
- **Radio Buttons:** The UI presents radio buttons for selecting different options. The available options depend on the file type. If an image is uploaded, the options are 'CGI-AI' and 'Copy-move/Splicing'. If a video is uploaded, the only option available is 'Deepfake'.
- **Submit Button:** A submit button is provided to trigger the processing and display of the results.

- **Result Display:** Once the submit button is clicked, the UI displays the uploaded image or video. The selected option determines the type of manipulation being checked. The result is then displayed in a larger font size with the heading 'RESULT'.

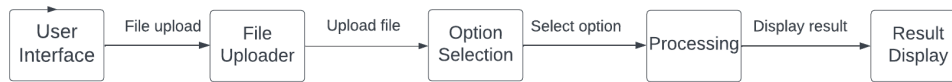


Figure 5.5: Block Diagram of UI

When a video is uploaded, the 'Deepfake' option is automatically selected and the radio buttons are disabled. After clicking the submit button, the 'Deepfake' function is called. On the other hand, if an image is uploaded and a radio button is selected, the corresponding function is called, which passes the video or image to the model and returns the result. The result is then displayed under the result section in the user interface.

Chapter 6

Results and Discussion

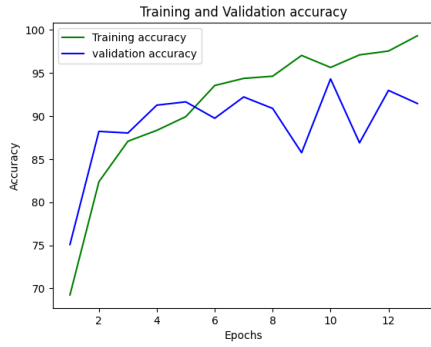
By integrating various models, each tailored to their respective fields, we developed an application capable of discerning between genuine and fabricated images and videos. The user-friendly interface framework streamlines the application's creation process while concealing its intricate inner workings. Below, we provide a comprehensive account of the individual models' results obtained during their training and evaluation phases.

6.1 Deepfake detector

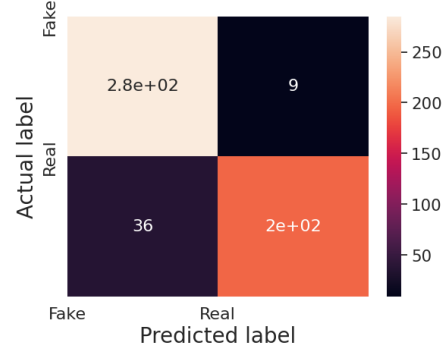
The model was trained and tested using the Celeb-DF [13] dataset. For training another face-only dataset was created from the Celeb-DF dataset. The dataset was split into 80% for training and 20% for testing. About 1577 videos were used for training out of which about 657 videos were real and 920 videos were fake. For testing about 526 videos were used out of which about 232 videos were real and 294 videos were fake.

The model was trained with a batch size of 4 and a learning rate of $1e-5$ (0.00001). The training was performed for 13 epochs. The cross-entropy loss was used as the loss function and the Adam optimizer with a weight decay of $1e-3$ is used to optimize the parameters.

The performance of the model was evaluated using the accuracy metric on a separate test set. The model achieved an accuracy of 91.44 and a loss of 0.26 on the test set after 13 epochs. The model didn't get any better. After that, the performance of the model begins to diverge.



(a) Model Accuracy



(b) Confusions Matrix

Figure 6.1: Deepfake detection model

6.2 Synthetic media detector

Our proposed model outperforms the Efficient-Net classifier, another image classification model, in terms of performance. When working with smaller batch size values and dataset sizes, our model may become less generalized. However, due to the pre-training of the base model, it exhibits strong performance from the initial epoch. Accuracy and loss values demonstrate comparable performance on both the training and testing sets. Subsequent to approximately 10 epochs, the improvements are incremental, and the learning rate is automatically adjusted for smaller step sizes to enable gradual progress. Around 20 epochs, the performance on the train and test sets begins to diverge. To ensure the model's optimal state during the epoch, early stopping was implemented. The final model shows around 91% accuracy and good precision and recall ability in the validation set which can be seen in Figure 6.2 The model gradually reaches a plateau, and continuing to train it results in either no improvement or minimal gains.

6.3 Manipulation detector

CASIA2 and NC16 datasets were used to train and test the model. Without data augmentation, patches were obtained from 2,204 and 20,076 images, respectively, of the NC16 and CASIA2 datasets.

The CNN model achieved an accuracy of 82.3 after 30 epochs, as shown in Fig 6.3a. Then the change in accuracy was slow and finally reached 93.5 after 250 epochs.

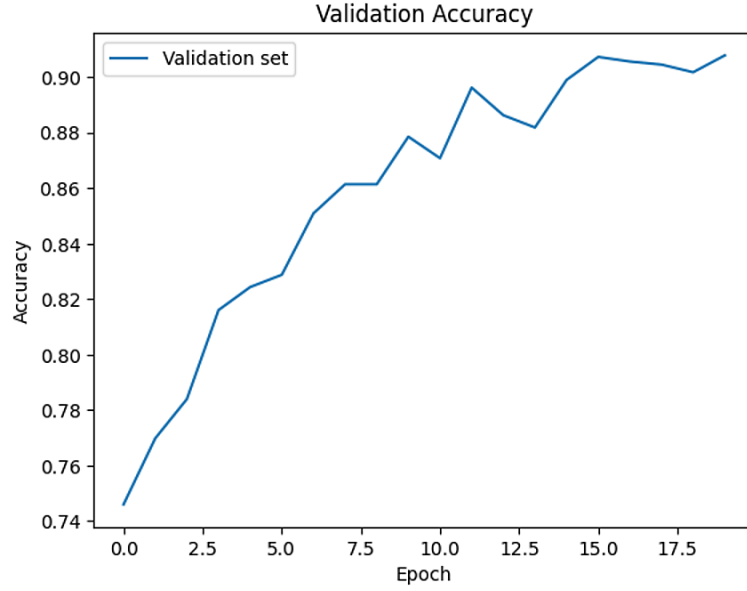
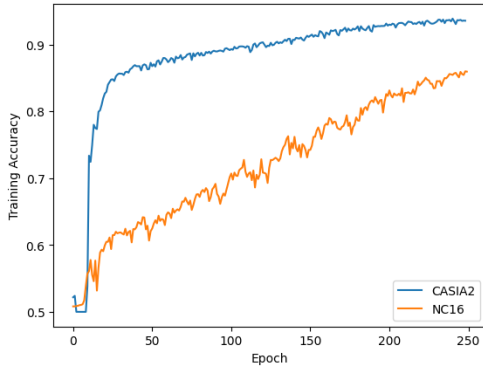
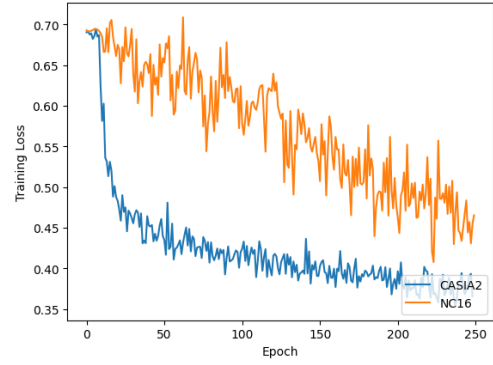


Figure 6.2: Synthetic media detector validation accuracy



(a) Accuracy on CASIA2 and NC16 datasets



(b) Loss on CASIA2 and NC16 datasets

Figure 6.3: Accuracy and loss of CNN model

The model's loss decreased steadily over the training period, reaching a final value of 0.365, as shown in Figure 6.3b.

SRM initialization filters were utilised in the first convolutional layer of the CNN, and the hyperparameters of the CNN were individually optimised for each dataset. The features generated from the CNN were combined using mean feature fusion, and an SVM model with an RBF kernel was employed for classification.

Using 10-fold cross-validation, the accuracy of the SVM model was assessed; it had an accuracy of 0.9682 and an error of 0.0119. Additionally, 1426 true negatives, 72 false negatives, 17 false positives, and 1008 true positives were reported (Fig. 6.4).

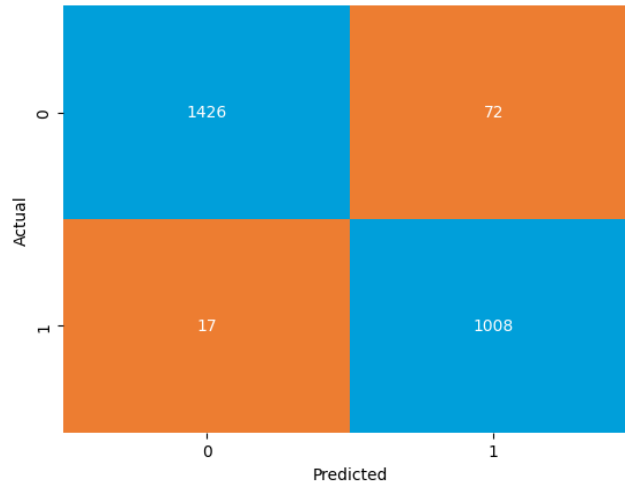


Figure 6.4: Confusion Matrix for SVM (0-pristine, 1-fake)

6.4 User interface

Our designed user interface provides the user with the ability to upload image and video files and select different options for detecting various forms of manipulation. The user is presented with a file uploader widget that enables them to select an image or video file. Depending on the type of file uploaded, relevant options are displayed on the radio buttons. The user can select an option by clicking on the corresponding radio button. A visual representation of the output can be observed in Figure.6.5.

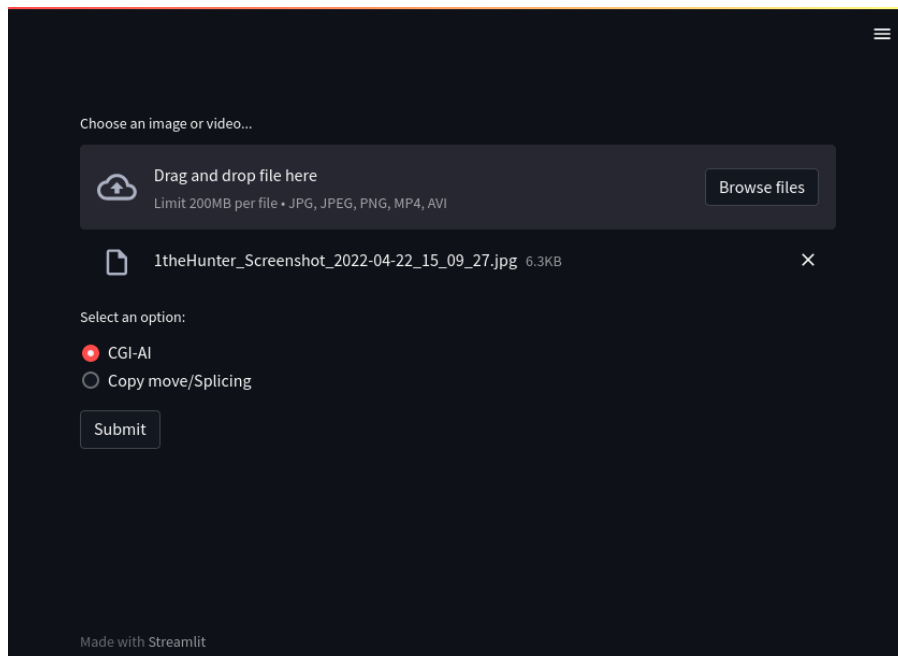


Figure 6.5: Sample output screenshot

Chapter 7

Conclusion

The project "Video and Image Manipulation Detection" focuses on using deep learning techniques to detect deepfakes, AI-generated, and computer-generated media, which are causing serious concerns because of their potential for misuse. These manipulated media can be used for malicious purposes, such as spreading false information or damaging someone's reputation.

The project aims to address this issue by developing reliable methods for detecting manipulations in visual media, with a particular emphasis on forensic analysis. The project explores and analyzes existing technologies for detecting deepfakes and CGI manipulations, and also develops and tests novel approaches to arrive at a robust and effective solution.

The project's performance evaluation demonstrates that the proposed methods are effective in detecting various types of manipulations. These methods can be useful for forensic analysis, journalism, social media, and other applications where the authenticity of visual media is essential. They can help ensure the credibility and trustworthiness of visual media in various contexts, and aid in investigations and legal proceedings.

In summary, the "Video and Image Manipulation Detection" project is a significant contribution to combat the growing problem of manipulated visual media. Its proposed deep learning techniques are promising for detecting manipulations and have the potential to make a significant impact on various applications where the authenticity of visual media is crucial.

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